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Multi-Metric Representation Learning Strategy Based on Clustering for Fine-Grained Multimodal Sentiment Analysis

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Abstract

*Multimodal sentiment analysis (MSA) aims to identify human emotions through multimodal data. Despite considerable advances in MSA, we find that emotional class centers often overlap when integrating data from different modalities into the same representation space. In this paper, we propose a novel **M**ulti-**M**etric **R**epresentation l**e**arning **s**trategy based on clus**t**ering (MMRest) to alleviate this issue through flexible multi-metric representation learning, enabling the model to learn fine-grained sentiments. Specifically, we first design a module termed Multi-metric Multimodal learning on Clusters (MMC), which minimizes distances within similar sentiment pairs while maximizing dissimilar ones, aiming to learn a global metric and local metrics in each cluster from multimodal data. Afterwards, we develop a Projection and Decision-Level Fusion (PDLF) module, including two parts. One part utilizes the optimal global and local metrics to obtain a projection value. The other part combines the projection value with an intermediate score which is obtained through the fusion of unimodal and multimodal representations to obtain the final sentiment prediction score. Extensive experiments on the CMU-MOSI and CMU-MOSEI datasets demonstrate that our method is significantly superior to state-of-the-art methods on various evaluation indicators and parameter count, by effectively learning fine-grained emotional boundaries. Code:* https://github.com/hurriedpi/MMRest

1. Introduction

Multimodal Sentiment Analysis (MSA) has emerged as a pivotal research area, aiming to achieve a more comprehensive and accurate understanding of human sentiment by integrating information from multiple modalities, including language, vision, and acoustics. By leveraging these complementary data streams, MSA models can capture nuances that

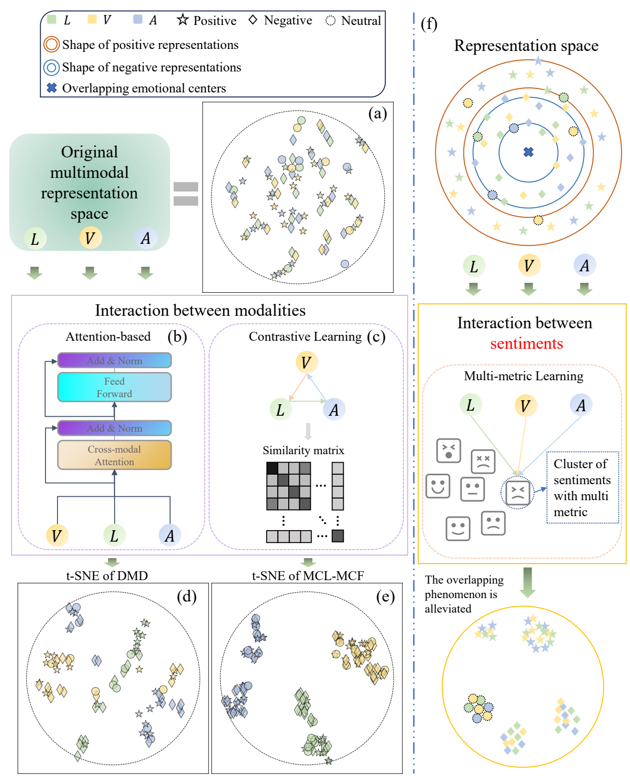

Figure 1. (a) denotes the original multimodal representation space suffering from the issue of overlapping sentiment class centers. Representations belonging to different sentiment categories can cluster closely together, blurring the decision boundaries. (b) and (c) illustrate two typical models that only focus on the interaction between modalities. (d) and (e) show two t-SNE visualization figures of DMD [17] and MCL-MCF [2], which reveal that the issue still exists, hindering the model's ability to make fine-grained distinctions. (f) represents our method, aiming to blur the influence of modality and further enhance the emotional semantic information of multimodal representations through multi-metric learning.

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are often overlooked by unimodal approaches, leading to advancements in fields such as human-computer interaction [12], opinion mining [21], and mental health analysis [37]. However, integrating data from heterogeneous modalities into the same representation space remains an open problem.

Under this background, researches on MSA can be roughly divided into two directions: multimodal fusion strategies [9, 18, 28, 31, 32, 44] and multimodal representation learning strategies [2, 4, 17, 35, 46]. As illustrated in Fig. 1(b), the former tends to utilize variants of cross-modal attention mechanisms [30], typically with text as the dominant modality for sufficient interaction between modalities, to achieve multimodal fusion. The latter primarily aims to achieve in-depth semantic comprehension of various modalities enriched with diverse human sentiment cues to develop more powerful encoders [31]. A typical method [2] of the latter is shown in Fig. 1(c), which uses multi-level contrastive learning to design different combinations of positive and negative samples for unimodal-unimodal pairs, unimodal-multimodal pairs, and multimodal-multimodal pairs, to learn semantically sufficient joint representations.

Although these methods have made significant progress in learning effective joint representations, we find that when integrating data from different modalities into the same representation space, sentiment class centers often overlap, as shown in Fig. 1(a). After DMD [17] and MCL-MCF [2] encoding multimodal representations, t-SNE [29] visualization of them represents that the issue indeed exists, as shown in Fig. 1(d) and Fig. 1(e). We argue that the training process of these models is greatly influenced by the modality. Although these models can effectively distinguish between modalities, their ability to identify different emotional centers is relatively weak. What is worse, both Transformer and contrastive learning incur high computational costs. There is a pressing need for a method that can directly model the intra-cluster (within the same sentiment) and inter-cluster (between different sentiments) relationships to pull representations of the same sentiment closer while pushing apart those of different sentiments, thereby alleviating the overlapping in the sentiment centers.

To this end, we propose a novel multi-metric representation learning strategy based on clustering (MMRest) to enable interaction between fine-grained sentiment representations using different metrics, as shown in Fig. 1(f). Specifically, we first introduce a module named Multi-metric Multimodal learning on Clusters (MMC) to achieve interaction between sentiments, in which the multimodal representations are projected into a common space and learn a global metric to obtain commonality in all sentiment clusters, and then learn a specific local metric for each sentiment cluster to capture distinctive characteristics. Then, a simple but effective Projection and Decision-Level Fusion (PDLF) module is designed to further boost interaction between modalities, which is divided into two branches. On the one hand, PDLF utilizes the multi-metric matrices and cluster centers learned in MMC to calculate the geometric projection value. On the other hand, PDLF interacts between unimodal and multimodal representations to predict an intermediate sentiment score. Finally, we integrate them to obtain the final sentiment prediction score. Our model is relatively lightweight compared to previous models [2, 8].

The contributions of this work can be summarized as:

- We draw an issue in MSA that emotional class centers often overlap when integrating data from different modalities into the same representation space. To alleviate this problem, we propose an MMRest model.
- In MMRest, we design a module termed multi-metric multimodal learning on clusters (MMC) to obtain the optimal global metric matrix corresponding to all fine-grained sentiments and the optimal local metric matrices for different fine-grained ones.
- A projection and decision-level fusion (PDLF) module is proposed, in which both the geometric projection value calculated by the optimal global and local metrics and intermediate sentiment prediction derived from the unimodal encoders are further integrated to obtain the final sentiment score.
- Extensive experiments on benchmark datasets demonstrate that our approach is lightweight while superior to existing state-of-the-art methods on various evaluation indicators. The visualization results and variance analysis also verify that the proposed multi-metric representation learning strategy is able to alleviate the issue we find.

2. Related Works

2.1. Multimodal Sentiment Analysis

Multimodal Sentiment Analysis (MSA) aims to understand human sentiments by integrating complementary information from various modalities, primarily language, visual, and acoustic signals. Early works in MSA often focused on designing sophisticated fusion strategies, such as tensor-based fusion [22, 41] and memory-augmented networks [42], to combine unimodal representations. Graph-based fusion methods [13, 23] are also studied, which explore the cross-modal associations across time series and modalities. Later, the powerful sequence modeling capability of Transformer [30] has spurred the development of interactive networks based on cross-modal attention [6, 28, 44, 45]. To capture the relationship between unimodal-unimodal and unimodal-multimodal representations pairs, recent studies [2, 36] designed complex contrastive learning frameworks for modal interaction. Nevertheless, these approaches often incur high computational costs and may still struggle to sufficiently resolve fine-grained emotional discrepancies under significant modality gaps. More recent approaches [15, 16, 19, 36, 48] have begun to address inherent challenges in multimodal learning, particularly the modality gap—the distributional discrepancy between representations of different modalities. Moving beyond traditional fusion, methods [3, 4] tackle inherent multimodal noise by quantifying unimodal uncertainty, thereby learning more robust representations. However, many existing methods struggle with the problem of sentiment center overlapping in the representation space, where representations of different fine-grained sentiments are insufficiently discriminated, limiting their performance on subtle sentiment distinctions. Our work tackles this challenge by learning sentiment cluster-aware representations that explicitly model intra-cluster compactness and inter-cluster separation.

2.2. Multi-Metric Learning

Multi-metric learning aims to simultaneously learn multiple local metrics to adapt to diverse data types and tasks, thereby providing more accurate and comprehensive measurements and capturing complex underlying structures in the data. Based on whether information is shared between local and global metrics, multi-metric learning methods can be categorized into two types: unshared and shared. The former refers to the models [11, 26, 33, 38, 47] learning only local metrics, without simultaneously learning global metrics. CDMML [33] used extracted facial features to learn multiple distance metrics and further utilized complementary and discriminative information for recognition. DMLCN [47] divided the sample into multiple clusters and learned different metrics based on different cluster centers to better adapt to complex data structures. The latter refers to the models [5, 20, 24, 27, 34] learning multiple local metrics while learning global metrics. Tan *et al*. [27] integrated metric learning and graph learning into multi-view clustering, and utilized filtering ideas to restore smooth representations of data by exploring its geometric structure and preserving the original geometric structure of the data. Yan *et al*. [34] adaptively generated samples with domain diversity and enhanced the generalization ability of the model. Compared to learning local metrics only, learning multiple metrics can better capture relationships within the data. Thus, exploiting multi-metric learning to capture the complex internal structure between multimodal data is a good choice. Inspired by the analysis of the above literature, we propose a shared multi-metric learning strategy for fine-grained multimodal sentiment analysis.

3. Methodology

In this section, we present a novel multimodal representation learning framework MMRest, for fine-grained multimodal sentiment analysis. The overall framework is illustrated in Fig. 2. Our approach aims to learn the relationships within and between fine-grained sentiment clusters through different metrics, thereby alleviating the problem of overlapping sentiment class centers in the representation space. What's more, without cross-modal attention mechanisms, the MMRest leads to a certain advantage in parameter count. MMRest contains two modules: the MMC module and the PDLF module. In the MMC module, we first perform feature extraction and alignment on multimodal inputs for subsequent clustering. After performing k-means algorithm, we employ a multi-metric learning strategy, combining a global metric and a set of local metrics for different sentiment clusters, to enhance the model's capacity for capturing fine-grained emotional boundaries. Finally, we design the PDLF module to utilize the metrics and cluster information learned from MMC to calculate the geometric projection value, which is then combined with the intermediate sentiment scores obtained from multimodal interactions.

3.1. Problem Definition

Given a batch of input samples with three modalities $X^m \in \mathbb{R}^{B\times d_m \times T_m}$, where $m \in \{L, V, A\}$ presents a modality, our goal is to predict a sentiment score $\hat{y}$ for each sample. T_m is the sequence length and d_m is the feature dimension.

3.2. MMC

As illustrated in Fig. 2(a), firstly, we process multimodal input samples X^m into unit-length representations $C^m \in \mathbb{R}^{B\times d_m}$ following previous works [2, 7]. Specifically, for visual and audio modalities, we adopt LSTM [10] to extract the final hidden state, *i.e*., C^V and C^A with global semantic information from sequences of visual and auditory inputs. For language modality, we utilize BERT [1] to encode input samples X^L and take the first token as the representation $\hat{C}^L$ of the language modality. In order to implement the clustering algorithm in the same dimensional multimodal representation space, we use a linear layer to align the dimension of the language modality with the other two modalities to d_1. Formally,

$$C^V = LSTM(X^V) \in \mathbb{R}^{B\times d_1}, \tag{1}$$

$$C^A = LSTM(X^A) \in \mathbb{R}^{B\times d_1}, \tag{2}$$

$$\hat{C}^L = BERT(X^L), C^L = Linear(\hat{C}^L) \in \mathbb{R}^{B\times d_1}. \tag{3}$$

In MSA, discrepancy between modalities could lead to the problem of overlapping sentiment centers, resulting in the representations of different sentiments being distributed around the same or very close center, as shown at the top of Fig. 1(e). In order to learn fine-grained emotional discrepancy and clear emotional boundaries, we propose a novel multi-metric representation learning strategy.

Specifically, taking two widely used datasets, CMU-MOSI [40] and CMU-MOSEI [43], as examples where the label values range from -3 to 3, we first divide the value range y into 7 intervals and assign a unique cluster label y^C

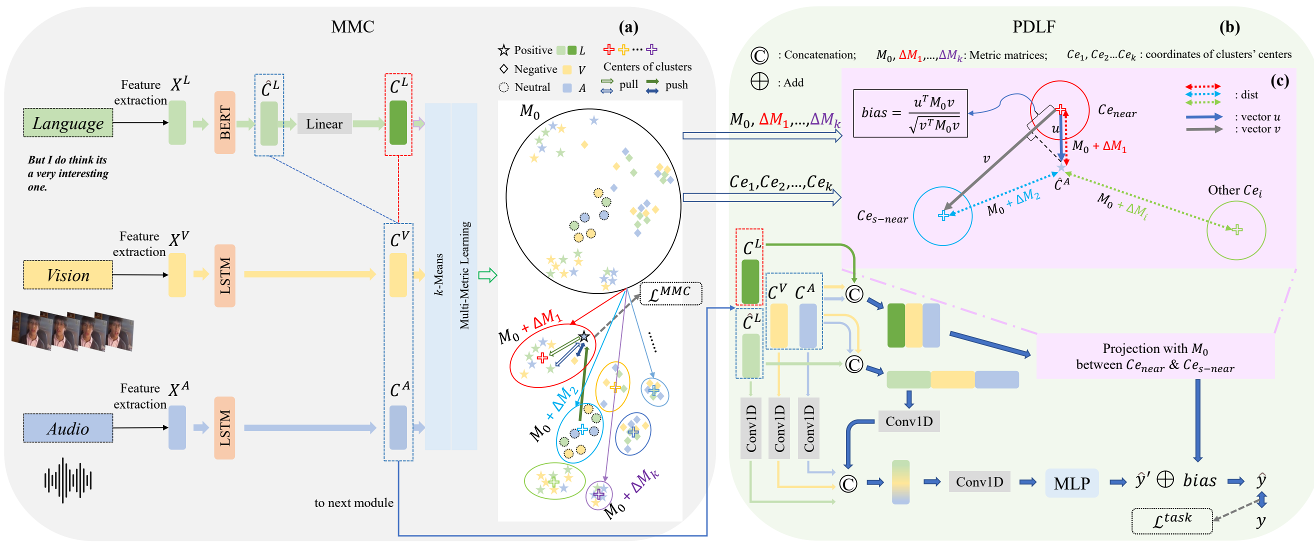


Figure 2. overall framework of MMRest

to each unimodal representation. The cluster label y^C for the interval [-3.0, -2.5] is 0, (-2.5, -1.5] is 1, . . . , [2.5, 3.0] is 6. Then, we use the k-means clustering algorithm to segment the representations in a common representation space, blurring modality information. We set k as the hyperparameter. After completing the k-means clustering algorithm, we obtain k clusters. The receptive field of the i-th cluster center is defined as:

$$R_i = \{C^m | \forall Ce_i, \|C^m - Ce_i\|_2^2 \leq \|C^m - Ce_j\|_2^2\}, \quad (4)$$

where $i \in \{1, 2, \ldots, k\}$, $Ce_i = \frac{1}{M}\sum_{j=1}^{M} C_j^m$, and M represents the amount of nearest representations around the cluster center Ce_i. Each receptive field R_i may contain representations from different modalities and sentiment categories.

Next, we assign cluster label $y^{C_{ei}}$ to each cluster based on a specialized voting mechanism. Specifically, first, we determine the cluster label based on the distribution ratio of the label value range of the samples in the receptive field. For example, when the proportion of samples with label values ranging from [-3, -2.5] in the receptive field of a certain cluster is the highest, then the label of this cluster is set to 0. Then, based on the receptive field R_i, similar pairs and dissimilar pairs are defined as:

$$\mathbf{S_i} = \{(p, q) | C_p^m \in R_i, C_q^m \in R_i \text{ and } {y^C}_p = {y^C}_q\}, \quad (5)$$

$$\mathbf{D_i} = \{(p, q) | C_p^m \in R_i, C_q^m \in R_i \text{ and } {y^C}_p \neq {y^C}_q\}. \quad (6)$$

Fine-grained sentiments exhibit both shared and distinctive characteristics. To better capture and leverage this interplay, we introduce a set of metric matrices to learn complex emotional representations with high flexibility, *i.e.*, we adopt a global metric M_0 for all clusters and a series of local metrics $\varDelta M_i$ for each cluster Ce_i, as illustrated in Fig. 2(a).

We treat each R_i as a fundamental unit for optimization, aiming to pull representations closer in the embedding space if they share a similar emotional value range, and push them apart otherwise. Specifically, for similar and dissimilar pairs in R_i, we use global metric M_0 and local metric $\varDelta M_i$ to calculate the distance between multimodal representations, and optimize the metric matrices using hinge loss. The specific formula is as follows:

$$\mathcal{L}_{\text{intra}}^{(i)} = \max\left(0, \sum_{(p,q)\in\mathbf{S_i}} d_{M_i}(C_p^m, C_q^m) - \sum_{(p,q)\in\mathbf{D_i}} d_{M_i}(C_p^m, C_q^m) + \xi\right), \quad (7)$$

where $M_i = M_0 + \varDelta M_i + \beta I$, $d_{M_i}(C_p^m, C_q^m) = (C_p^m - C_q^m)^\top M_i (C_p^m - C_q^m)$ and ξ is a hyperparameter, I represents the identity matrix, $\beta > 0$ aims to take care of possible ill-conditioning of $M_0 + \varDelta M_i$. We add the intra-class hinge losses of each cluster to obtain the total loss $\mathcal{L}_{\text{total-intra}}$, formally:

$$\mathcal{L}_{\text{total-intra}} = \sum_{i=1}^{k} \mathcal{L}_{\text{intra}}^{(i)}. \quad (8)$$

We define triplet combinations between representations C_n^m and cluster centers as $\mathcal{T}_{ij} = \{(n, i, j) | {y^C}_n = y^{C_{ei}} \text{ and } {y^C}_n \neq y^{C_{ej}}\}$. To ensure the distance between different receptive fields is sufficiently far, our strategy is to separate the distance between C_n^m and cluster centers by at least ρ_{ij}.

$$d_{M_j}(C_n^m, Ce_j) - d_{M_i}(C_n^m, Ce_i) \geq \rho_{ij}, \quad (9)$$

where $i \neq j$, $\rho_{ij} = \alpha \|Ce_j - Ce_i\|_2^2$, α is a hyperparameter to control the weight of the distance between cluster centers.

Eq. (9) can be regarded as a variant of triplet loss, so this formula can be written as:

$$\mathcal{L}_{\text{inter}}^{(ij)} = \sum_{n \in \mathcal{T}_{ij}} \max\big(0, d_{M_i}(C_n^m, Ce_i) - d_{M_j}(C_n^m, Ce_j) + \rho_{ij}\big)\,, \tag{10}$$

$$\mathcal{L}_{\text{total-inter}} = \sum_{i=1}^{k} \sum_{j=1, j \neq i}^{k} \mathcal{L}_{\text{inter}}^{(ij)}. \tag{11}$$

Finally, we integrate the losses of the MMC module to obtain the overall loss function:

$$\mathcal{L}_{\text{MMC}} = \mathcal{L}_{\text{total-intra}} + v_1 \mathcal{L}_{\text{total-inter}} + v_2 \left\| M_0 \right\|_F^2\,, \tag{12}$$

where v_1, v_2 are hyperparameters, $\|\cdot\|_F^2$ indicates the square of the Frobenius norm of a matrix as a regularization term.

Optimization algorithm. Directly using the traditional Adam [14] optimizer may lead to metric matrices M_0 and ΔM_i being non-positive definite. Thus, we perform eigen-decomposition, forcing eigenvalues greater than or equal to 0. Specifically, for M_0, in the $t\text{-}th$ time step, we perform the following operations and the updating of ΔM_i is the same as M_0:

$$M_0^{(t+\frac{1}{2})} = \frac{1}{2}(Adam(M_0^{(t)}) + Adam(M_0^{(t)})^\top), \tag{13}$$

$$Q \Lambda Q^\top = Eig(M_0^{(t+\frac{1}{2})}), \tag{14}$$

$$M_0^{(t+1)} = Q \operatorname{diag}\left(\max(0, \lambda_1), \ldots, \max(0, \lambda_{d_1})\right) Q^\top, \tag{15}$$

where $Adam(\cdot)$ indicates the parameter update process of traditional Adam algorithm, $Eig(\cdot)$ represents eigendecomposition operation, $\lambda_1, \ldots, \lambda_{d_1}$ are the elements on the diagonal of Λ.

3.3. PDLF

In the PDLF module, we integrate emotional semantic information learned in MMC with the results of interaction between unimodal and multimodal representations to obtain final sentiment prediction scores. As illustrated in Fig. 2(b), we calculate the sentiment prediction scores for two parts:

Part I. We concatenate the tri-modal representations, *i.e.*, C^m, for batch computing, as shown in Fig. 2(c). Firstly, the Mahalanobis distance between these integrated representations and every cluster center is computed under the respective metric M_i:

$$C^f = \text{concatenate}(C^L, C^V, C^A) \in \mathbb{R}^{3B \times d_1}, \tag{16}$$

$$diff\_Ce_i = C^{f\ (3B \times d_1)} - Ce_i^{(1 \times d_1)} \in \mathbb{R}^{3B \times d_1}, \tag{17}$$

$$dist\_Ce_i = \sum_1^{d_1} (diff\_Ce_i \cdot M_i) \odot diff\_Ce_i \in \mathbb{R}^{3B}, \tag{18}$$

$$dists = \text{concatenate}(dist\_Ce_1, \ldots, dist\_Ce_k) \in \mathbb{R}^{3B \times k}, \tag{19}$$

where $\odot$ is Hadamard product, $M_i = M_0 + \Delta M_i + \beta I$ and Eq. (17) adopts a broadcast mechanism. Then, by searching the nearest and sub-nearest cluster centers for each representation, *i.e.*, $Ce_{near} \in \mathbb{R}^{3B \times d_1}$ and $Ce_{s-near} \in \mathbb{R}^{3B \times d_1}$ according to Eq. (19), we compute $bias$ aiming to achieve more accurate prediction of sentiment scores, formally:

$$bias_{multi} = \frac{\sum_1^{d_1} (\mathbf{u} \cdot M_0) \odot \mathbf{u}}{\sqrt{\sum_1^{d_1} (\mathbf{v} \cdot M_0) \odot \mathbf{v}}} \in \mathbb{R}^{3B}, \tag{20}$$

$$bias^L, bias^V, bias^A = \text{chunk}(bias_{multi}) \in \mathbb{R}^B, \tag{21}$$

$$bias = \frac{1}{3}(bias^L + bias^V + bias^A), \tag{22}$$

where $\mathbf{u} = C^f - Ce_{near}$ and $\mathbf{v} = Ce_{s-near} - Ce_{near}$. In the $bias$, projection values are obtained from the multimodal representation space, which is well clustered by the MMC module according to sentiment.

Part II. As illustrated in the lower part of Fig. 2(b), we aim to combine the $bias$ derived by **Part I**, which contains multimodal representation information under the optimal metrics obtained by MMC with the unimodal representations to make the final sentiment predictions. First, we perform Conv1D on the unimodal representations and then concatenate them as $[\hat{C}^L, C^V, C^A]$ to unify the dimensions to d_2. Then, we concatenate these four intermediate representations and perform another Conv1D before sending them to MLP to output the predicted score $\hat{y}'$. After adding scores of the two parts, we obtain the final sentiment score $\hat{y}$ for downstream tasks. Formally,

$$\hat{y} = \hat{y}' + \gamma bias, \tag{23}$$

where γ is a hyperparameter. Following previous works [2, 31, 32], we use L1 loss as the task loss. The final complete loss is as follows:

$$\mathcal{L} = \eta \mathcal{L}_{\text{MMC}} + \mathcal{L}_{\text{task}}. \tag{24}$$

Motivation and design principle of $bias$. Actually, after the MMC module, we can get the optimal M_i for each sentiment receptive field and optimal M_0 for global metric. Of course, we can apply such metrics to make the final sentiment prediction. In this paper, we believe that putting all multimodal data together, and emotional clustering will lose the difference between single modalities. Therefore, we attempt to link the optimal metrics obtained by the MMC module with the data of single modalities outputs by the PDLF module. The most common idea is to map the MMC module and PDLF module to the same representation space, but this is very difficult because we need to decompose M_i into $U^\top U$, and the optimal representation for C^f can be rewritten as $C^f U$. This is obviously difficult and will cause additional complexity and information loss in the process of matrix decomposition.

Our idea is to combine the optimal M_i in the form of $bias$ with the sentiment prediction scores of the unimodal data of the PDLF module. In terms of geometric structure, different emotional categories should have a low-dimensional manifold structure, *i.e.*, after clustering, the similar sentiment categories will be close to each other in low-dimensional representation space, while the dissimilar sentiment categories will be far away. Therefore, we first select the two class centers of the nearest and the sub-nearest for each C^f, and then project them in the form of Eq. (20). This geometric projection method not only has a good interpretation, but also avoids the use of cross-attention mechanisms and further reduces the parameter count of the model.

4. Experiments

In this section, we conduct a series of experiments to comprehensively evaluate our MMRest, including comparison experiments, ablation study, efficiency study, and visualization study. Due to space limitations, variance analysis, case study and hyperparameter study can be found in the supplementary material.

Datasets. We conduct experimental verification on the widely used multimodal sentiment analysis benchmark datasets CMU-MOSI [40] and CMU-MOSEI [43]. **CMU-MOSI** dataset contains 2199 monologue video clips extracted from 93 film review videos. Each segment is labeled with a continuous value of emotional polarity within the interval [-3, 3], representing *extreme negativity, negativity, mild negativity, neutrality, mild positivity, positivity, and extreme positivity*. The dataset is divided into 1284 training samples, 229 validation samples, and 686 test samples according to general partitioning criteria. **CMU-MOSEI** is an extended version of CMU-MOSI, which is larger in scale and more challenging. It contains 22856 clips from YouTube monologue videos, with emotional annotations also distributed in the continuous space of [-3, 3]. The official classification of the dataset includes 16326 training samples, 1871 validation samples, and 4659 test samples.

Evaluation Indicators. Mean Absolute Error (MAE), Pearson Correlation Coefficient (Corr), 7-class accuracy (Acc-7), 5-class accuracy (Acc-5), 3-class accuracy (Acc-3), binary accuracy (Acc-2), as well as F1 score, where binary classification and F1 score evaluation are reported separately according to two classification methods (negative/non-negative, negative/positive).

Implement Details. In all experiments, we utilize unaligned data provided by the open source [7]. All experiments, including reproducing results, are accomplished using PyTorch on an RTX 4070tisuper GPU with 16GB memory. Batch size of CMU-MOSI and CMU-MOSEI are set as 64 and 128, respectively. More experimental settings can be found in the supplementary material. To ensure the convergence of the model, we adopt early stopping with a patience of 10 epochs.

4.1. Comparison experiments

We compare our model with the following 11 baselines: TFN [41], LMF [22], MulT [28], MAG-BERT [25], MISA [8], Self-MM [39], MMIM [7], DMD [17], DEVA [32], DLF [31], MCL-MCF [2].

Tab. 1 illustrates the comparison on CMU-MOSI and CMU-MOSEI datasets. To minimize the influence of the experimental environment, we reproduced 5 recent open-source methods under the same conditions, using the code and optimal parameters provided in the original papers. Our method achieves optimal or suboptimal performance in all evaluation indicators on these two datasets. On the CMU-MOSI dataset, our method achieves 1.32% and 1.46% improvement on Acc-5 and Acc-7, respectively, compared to the suboptimal results in MCL-MCF [2]. We speculate that, on the smaller dataset, the huge network structure adopted by the multi-level contrast learning method in MCL-MCF [2] tends to pay too much attention to the interaction between modalities, thus learning a lot of redundant information. MMRest can effectively learn multimodal representations of emotional information, thereby improving the ability to recognize fine-grained sentiments. Although our model does not perform as well on the larger CMU-MOSEI dataset as on the smaller CMU-MOSI dataset, it still improves by 0.75% on Acc-2. In addition, our MMC module and PDLF module are lightweight, referring to our further analysis on the model parameter count in Sec. 4.3.

4.2. Ablation study

Ablation studies on MMRest are conducted from the following three strategies:

Modalities. As shown in Tab. 2, we first ablate the three modal representations. The results illustrate that when only text input exists, the model still has excellent sentiment prediction ability in MSA. When the audio or visual modality is retained only, the model cannot be trained properly. However, when we fuse the three modalities and train them together, the model shows significant improvement, with the highest improvement being 4.16%. This also indicates that the design of the PDLF module is necessary to facilitate interaction between different modalities.

Key Modules. Compared to complete MMRest, for MMRest(*w/o* $\mathcal{L}_{\text{MMC}}$, *w/o* $bias$), the performance of Acc-5 and Acc-7 on the CMU-MOSI dataset decreases significantly, by 4.82% and 3.79% respectively. MMRest(*w/* $\mathcal{L}_{\text{MMC}}$, *w/o* $bias$) exhibits a marginal overall performance drop as well.

*https://github.com/thuiar/MMSA/blob/master/results/result-stat.md

Table 1. Comparison on CMU-MOSI and CMU-MOSEI datasets.

Models	CMU-MOSI						CMU-MOSEI					
	Acc-2↑	Acc-5↑	Acc-7↑	F1↑	MAE↓	Corr↑	Acc-2↑	Acc-5↑	Acc-7↑	F1↑	MAE↓	Corr↑
TFN*	77.99/79.08	-	34.46	77.95/79.11	0.947	0.673	78.50/81.89	-	51.6	78.96/81.74	0.572	0.714
LMF*	77.90/79.18	-	33.82	77.80/79.15	0.950	0.651	80.54/83.48	-	51.59	80.94/83.36	0.575	0.716
MulT*	79.71/80.98	42.68	36.91	79.63/80.95	0.879	0.702	81.15/84.63	54.18	52.84	81.56/84.52	0.559	0.733
MAG-BERT	82.37/84.43	-	43.62	82.50/84.61	0.727	0.781	82.51/84.82	-	52.67	82.77/84.71	0.543	0.755
MISA*	81.84/83.54	47.08	41.37	81.82/83.58	0.776	0.778	80.67/84.67	53.63	52.05	81.12/84.66	0.557	0.751
Self-MM†	82.89/84.63	52.32	45.99	82.93/84.71	0.714	0.793	82.69/84.86	54.89	53.48	82.73/84.97	0.535	0.761
MMIM†	81.49/83.69	51.90	45.48	81.30/83.59	0.726	0.785	81.78/83.82	51.81	50.38	81.94/83.56	0.550	0.759
DMD†	82.51/83.84	53.21	46.06	82.51/83.89	0.749	0.788	79.95/84.26	54.69	53.02	80.58/84.28	0.544	0.755
DEVA	84.40/86.29	51.78	46.32	84.48/86.30	0.730	0.787	83.26/86.13	55.32	52.26	82.93/86.21	0.541	**0.769**
DLF†	81.92/83.84	52.04	44.46	81.92/83.84	0.733	0.784	80.83/84.70	54.41	52.52	81.37/84.69	0.544	0.759
MCL-MCF†	82.22/84.30	55.10	47.52	82.17/84.32	0.693	**0.807**	78.97/84.70	55.93	54.06	79.79/84.82	0.539	0.759
Ours	**84.64/86.78**	**56.42**	**48.98**	**84.75/86.89**	**0.683**	0.802	**83.42/86.88**	**56.52**	**54.72**	**83.65/86.89**	**0.531**	**0.769**

* represents results obtained from and its corresponding GitHub page*. Models with † are reproduced under the same conditions. Best results are marked in bold, and suboptimal results are marked with an underline.

The most significant declines are observed on Acc-5 and Acc-7, which decrease by 2.34% and 1.46%, respectively. On the larger CMU-MOSEI dataset, the results of ablation studies are still consistent with the conclusions drawn on the CMU-MOSI dataset. These results indicate that the proposed $\mathcal{L}_{\text{MMC}}$ loss enhances the representation space structure by enforcing intra-cluster compactness and inter-cluster separation among representations with fine-grained sentiments, achieving the best performance by employing optimal global and local metrics to compute the geometric projection value for the final sentiment score. Furthermore, the MMC module, in conjunction with the PDLF module, is able to effectively optimize our model and significantly improve its ability to learn fine-grained sentiments.

Metrics. The MMC module of our model adopts two metrics: the global metric M_0 and the local ΔM_i for fine-grained sentiments. We conduct an ablation study to evaluate the effectiveness of the two types of metrics in the proposed multi-metric learning strategy. The results on CMU-MOSI dataset are presented in Tab. 3. Replacing the learnable global metric matrix M_0 and local metric matrices ΔM_i (*i.e.*, using multi-metric) with identity matrices (*i.e.*, using single-metric) results in a slight performance drop on Acc-2 and Corr, but a sharp decline on Acc-5, Acc-7, and MAE. This significant performance degradation confirms that our multi-metric learning strategy is crucial for modeling fine-grained sentiment information.

4.3. Efficiency Study

In this section, we conduct an efficiency study, comparing the proposed MMRest with prior Transformer-based methods: MulT [28], MAG-BERT [25], MISA [8], and the contrastive learning-based method MCL-MCF [2] in terms of parameter size and performance on CMU-MOSI.

As shown in Tab. 4, the parameter size of our model is only approximately 30% of the parameter size of the lightest Transformer-based model: MAG-BERT. However, our performance of Acc-5 and Acc-7 greatly exceeds these models, indicating that Transformer-based methods may learn redundant information and lead to suboptimal performance.

MCL-MCF uses multi-level contrastive learning to optimize different encoders, where its core contrastive learning part generates three large-scale networks, resulting in a sharp increase in parameter size. In addition, GPU memory usage and running time are close to two to three times that of MMRest. Despite the complex network adopted by MCL-MCF, our model still outperforms it on both datasets.

4.4. Visualization

We provide t-SNE [29] visualization images of multimodal representations and fused representations in MMRest, and compare them with MCL-MCF [2], as illustrated in Fig. 3.

Fig. 3(a) and Fig. 3(b) show that: in unimodal representations, only language clearly distinguishes sentiments, while the others overlap to varying degrees. Furthermore, only from the subtle details, the aggregation effect of MMRest's multimodal representations is better than that of MCL-MCF. Therefore, we further compare the fused representations in Fig. 3(c) and Fig. 3(d). It can be observed that although both models can distinguish fine-grained sentiments, our model yields not only clearer decision boundaries but also a more compact intra-class structure and better inter-class distances in the representation space.

To further validate the advantages observed in the visualizations, we design a measurement formula to measure the

Table 2. Ablation study on CMU-MOSI and CMU-MOSEI datasets. Best results are marked in bold.

Methods					CMU-MOSI					CMU-MOSEI				
L	V	A	$\mathcal{L}_{\text{MMC}}$	$bias$	Acc-2↑	Acc-5↑	Acc-7↑	MAE↓	Corr↑	Acc-2↑	Acc-5↑	Acc-7↑	MAE↓	Corr↑
✓			✓	✓	80.68/82.62	53.35	46.35	0.719	0.783	83.39/85.20	55.41	53.60	0.539	0.761
	✓		✓	✓	52.04/50.00	15.16	15.16	1.449	0.131	69.09/64.86	42.65	42.65	0.817	0.206
		✓	✓	✓	54.96/54.73	18.51	18.51	1.407	0.240	70.53/64.78	40.97	40.97	0.820	0.223
✓	✓	✓			83.53/85.21	51.60	45.19	0.711	0.795	81.48/85.03	55.76	53.74	0.535	0.762
✓	✓	✓	✓		84.02/86.12	54.08	47.52	0.696	0.797	81.98/85.23	55.69	53.98	0.547	0.746
✓	✓	✓	✓	✓	**84.64/86.78**	**56.42**	**48.98**	**0.683**	**0.802**	**83.42/86.88**	**56.52**	**54.72**	**0.531**	**0.769**

Table 3. Ablation study of metric methods on CMU-MOSI. Best results are marked in bold.

Methods	Acc-2↑	Acc-5↑	Acc-7↑	MAE↓	Corr↑
SM	80.76/83.23	35.42	32.94	0.914	0.713
MM	**84.64/86.78**	**56.42**	**48.98**	**0.683**	**0.802**

SM represents that MMRest uses a single-metric matrix, *i.e.*, the Euclidean distance metric matrix, MM represents that the model uses multi-metric matrices, *i.e.*, the M_0, ΔM_i distance metric matrices.

degree of overlap between centers of different classes:

$$\frac{\sum_i \sum_{j\neq i} \|\mathcal{C}_i - \mathcal{C}_j\|}{\sum_i \sum_{j\neq i} \left(\|\mathcal{C}_i - \mathcal{X}^{\mathcal{C}_i}_{\text{farthest}}\| + \|\mathcal{C}_j - \mathcal{X}^{\mathcal{C}_j}_{\text{farthest}}\|\right)}, \tag{25}$$

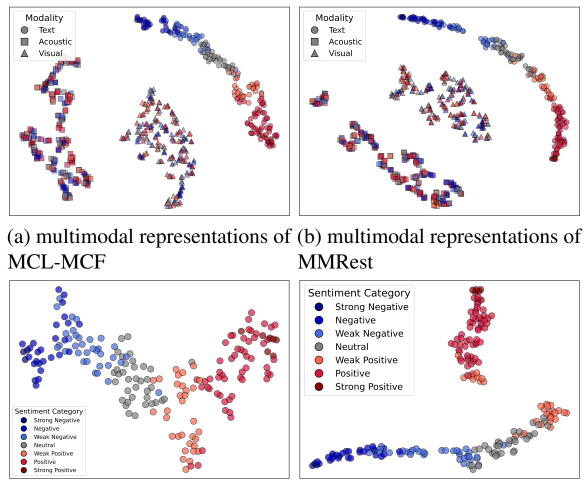
where $\mathcal{C}_i$ is the center of class i, and $\mathcal{X}^{\mathcal{C}_i}_{\text{farthest}}$ is the farthest representation from center $\mathcal{C}_i$. Values closer to 0 signal a more severe overlap of class centers. We average 10 batches of raw data, the result is 0.0719, which proves the overlap issue. We compare the results of existing models and MMRest in Tab. 5, and the results indicate that our MMRest can effectively alleviate the overlap issue.

Table 4. Efficiency study on CMU-MOSI dataset.

Models	Parameters↓	Memory↓	Time↓	Acc-5	Acc-7
MulT	2.57M	-	-	42.68	36.91
MAG-BERT	1.22M	-	-	-	43.62
MISA	3.10M	-	-	47.08	41.37
MCL-MCF	199.30M	11.1GB	4.26min	55.10	47.52
MMRest	**0.37M**	**4.4GB**	**2.46min**	**56.42**	**48.98**

Table 5. A quantitative analysis of the overlap issue on the CMU-MOSI dataset.

Methods	Raw data	MMRest	MCL-MCF	DMD
Results	0.0719	**2.0465**	1.2654	0.2870



(a) multimodal representations of MCL-MCF (b) multimodal representations of MMRest

(c) fused representations of MCL-MCF (d) fused representations of MMRest

Figure 3. t-SNE visualization of MCL-MCF and MMRest on CMU-MOSI dataset.

5. Conclusion and discussion

In this paper, we propose an MMRest model inspired by the issue we observed, that sentiment class centers often overlap when integrating data from different modalities into the same representation space. By integrating global and local metrics within a clustering framework, the proposed MMC module effectively models both intra-cluster and inter-cluster relationships, alleviating the issue. In addition, the PDLF module not only achieves the fusion of unimodal and multimodal representations, but also appropriately combines the optimal multi-metric matrices learned from the MMC module with the multimodal representations to calculate projection values, achieving more accurate fine-grained sentiment prediction. Extensive experiments on benchmark datasets demonstrate that our method outperforms state-of-the-art approaches in both performance and parameter count. One limitation is that MMRest cannot dynamically determine the number of clusters based on the size of the dataset. We will explore this in our future work.

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